Phase I: Reflection on the Use of GenAI to create two Model Behavior

The first chart developed from the threshold trade off graph is where it describes the false positive rate and false negative rate across different decision thresholds of MHS and ToxiGen. The second graph shows how the graph score versus human rating changes. This graph compares the model score with the human rating. It also identifies the correct predictions, false alarm and missed hate.
The key takeaway from both charts is that the end graphs were not created from one perfect prompt. Initially, the interaction started with analytical questions that later turned into specific ones. First, the request was essentially to find some useful way to study model behavior like a black box. Several possibilities were generated for visuals such as threshold analysis, confusion matrix, score distribution and rating comparison. That helped to open possibilities up, but it did not go far enough to pin down the exact graph wanted.
Due to the nature of prompts and the variables involved, the neural networks often make hidden assumptions. Just asking to generate “a graph comparing the datasets” is not enough. It should state what to compare, what each visual channel means, how to transform the data and what the viewer should be able to conclude from the figure.
GenAI proved to be especially helpful in transforming an analytical concept into working code. It quickly handled the basic parts of both graphs, loading the datasets, selecting the relevant columns, grouping the data, calculating the error categories, and creating plotting objects.

The graph of score versus rating was complicated, GenAI helped organize it. The figure is set up like this. The points show the prediction status of the model. The black line shows the average model score across the rating or score ranges. Finally, the vertical and horizontal dashed lines show the human rating and model score threshold respectively.
The initial versions were effective but not yet suitable as final figures. The first static graph did not make it easy to tell what was most important in the threshold graph. The text in the legend and on the lines were not clear indicators of whether something was for the dataset or error type. Based on my feedback we decided that color should always identify the dataset, while solid versus dashed lines should identify FPR versus FNR.
A major limitation is that even untested assumptions can still produce code that looks right. A graph can be technically accurate but misleading for the reader if, for example, the labels are reversed, the meaning of the score is misunderstood, or the two datasets shown are not comparable in the way the figure suggests. Also, the distribution of the score versus rating graph can show how close the errors are to the rating boundary or how far they are from the rating boundary.

Using GenAI speeds up the workflow, especially in the early stages. This helped to generate a good starting point, suggest alternatives, and turn feedback into code changes. By minimizing the need to search for plotting syntax, this helped in trying out different visual styles.

But, it slowed things down a bit. The first responses were wider and included extra plots that were not actually used. Although not a major headache, this shows how GenAI tends to do too much in response to an open ended prompt.

My main conclusion is that GenAI speeds up visualization work, but it does not replace the need for human judgment. Research questions, meaning of the data, visual encodings that are honest and understandable, and checking the final graph with the real datasets all have to be done by the human. While GenAI helped move the project along, the final graphs came down to human choices about what behavior really mattered to show and how viewers should read it.